

Agentic AI-The Killer Use Case for Blockchain and Crypto

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To capture the AI growth opportunity today, most investors buy shares of AI-aligned companies and related verticals. But will the same playbook work for agentic AI? See why cryptocurrencies and alt coins may be key to emerging agentic AI growth opportunities.

AI Evolution and Investment Dominance

AI continues to evolve. Early offerings in the 2010s around machine learning, natural language processing, and predictive analytics helped ignite the Big Data era and allow for the ingestion and processing of both structured and unstructured data at previously unimaginable speeds and volumes. AI was a tool that helped augment human work.

The release of generative AI capabilities in the early 2020s marked a significant expansion in its utility and role. AI became a co-creator, helping shape and respond to inquiries and take on aspects of human work. The full potential of generative AI is still unfolding as offerings improve and become integrated into ever-more aspects of daily life.

As its influence grows, the importance of AI as an investment driver is undeniable. On July 14, 2026, International Business Machines' (IBM) shares plunged 25.2% after it issued a warning that corporate technology spending was shifting increasingly to AI infrastructure, delaying or reducing spending on conventional software and IT projects.¹

Today, the S&P 500 is more concentrated than at any point since the late-1990s tech bubble. The 10 largest stocks are all AI-aligned and now make up almost 40% of the index's total market capitalization, compared to just 25% during the dotcom era and 15% in 1980.² Institutional investors in particular view AI as a structural megatrend as they allocate capital heavily to AI infrastructure, data centers and semiconductor stocks.

Yet, that investor positioning may prove inadequate to capture the potential of AI's next evolution.

The Emergence of Agentic AI

Just as generative AI marked a step change in power, effectiveness and use cases when compared to early AI tools, the growing maturation and adoption of agentic AI is likely to have an equally if not more profound impact on day-to-day life. Agentic AI advances the engagement model "from a reactive, conversational chatbot to an autonomous system that can perceive its environment, devise a plan and execute multi-step tasks to achieve high-level goals without constant human supervision."³

Agents can interact directly with external software systems and code repositories. This is recasting the role of AI. By 2028, 38% of organizations report that they will have AI agents as team members alongside humans, collaborating on driving productivity and innovation.⁴

In line with this advancement, the types of tasks that get off-loaded to AI are likely to become more complex. Generative AI offerings have excelled at knowledge-gathering and composition. AI agents are likely to take on growing numbers of transactions-initiating, tracking and fulfilling tasks independently and managing the outcomes. Estimates put agentic commerce as high as US\$3 trillion-US\$5 trillion by 2030.⁵

For institutions, such transactions are likely to occur within enterprise software applications, with forecasts showing that by 2028, 33% of offerings will include agentic AI and up to 15% of day-to-day decisions will be handled by such agents.⁶ Micropayments between software systems for compute resources, API calls, data utilization, and services represent a new interaction model that allows for task-level accounting and fulfillment.

Protocols are emerging that enable such machine-to-machine transactions. Stripe and Visa have rolled out the Machine Payments Protocol (MPP). Open-source solutions are also gaining traction. In the early 1990s, when the World Wide Web's architects wrote the rules for how browsers and servers communicated, they set aside a response code numbered "402" and labeled it "payment required". Coinbase created an "x402" protocol to enable agents to initiate and fulfill such instructions and then transferred their IP to the Linux Foundation, making it an open industry standard. Every

major credit card network, Web2 leaders like Stripe, Shopify, Google, and Amazon Web Services, and a growing set of Web3 providers have signed on to this payment standard so that "software can pay software" without human involvement.⁷

Agentic payments are likely to transform consumer engagement in the coming years. AI agents are expected to account for 15% to 25% of all US e-commerce sales by 2030.⁸ Already, ChatGPT processes 2.5 billion prompts daily with 53 million shopping queries flowing through AI platforms.⁹ OpenAI is also facilitating these checkout experiences within third-party ChatGPT apps (like Target, DoorDash and Instacart).¹⁰

Blockchain's Facilitation of Machine-to-Machine Transactions

Enabling these machine-to-machine transactions requires secure, autonomous, verifiable and high throughput recordkeeping. Legacy credit card and banking infrastructures are unsuited for agentic micropayments due to their fee structures; a standard credit card transaction averages 2%-3% plus a flat fee of approximately US\$0.30, versus AI agent payments that average US\$0.001 to purchase a single second of compute or a data query.¹¹

Agentic AI will likely need to rely on crypto technologies and blockchains to enable their activities as these rails are ideally suited for these use cases. Indeed, blockchains and crypto technologies are likely to become the foundational delivery layer for these transactions based on the following attributes:

Autonomous contract creation/fulfillment: AI payment agents generate tokens to make purchases and settle transactions. Each token embeds a set of transaction rules that set the parameters for the transaction, including the merchants that can accept the token, the maximum spending limit per transaction, and the timeframe for which the token is valid. These single-use tokens are automatically invalidated once a specific purchase is completed. Blockchains can hold, send, and receive these tokens and abide by the transaction rules just like they do for smart contracts.

Decentralized identity verification: AI agents have unique cryptographically verifiable identities. Each token they generate contains the agent's credentials that entitle it to sign blockchain-based transactions. Blockchains confirm these credentials as part of the transaction verification process and their consensus mechanisms prevent transactions from going through if the ID is not deemed legitimate.

Auditability: Every decision, transaction, or data exchange processed by an AI agent on blockchain can be recorded on the immutable ledger and be publicly recalled using block explorers to ensure accountability and transparency.

Access to decentralized computing and data: Blockchains enable agents to access distributed computing resources (e.g., GPU networks) and data, reducing reliance on centralized, proprietary cloud infrastructure and helping limit the cost of running a transaction-intensive model

Speed and fulfillment: While bitcoin only processes ~seven transactions per second and Ethereum ~75 transactions, newer high-speed chains are recording maximum speeds ranging from 12,933 transactions per second (TPS) on the Aptos chain, 6,284 TPS on Solana and 3,252 TPS on BNB Chain.¹² These transaction speeds are on par with the Visa network, which processes 1,700 to 10,000 transactions per second in normal operations.¹³ Yet, even this comparison is misleading. Blockchains both record and settle their transactions in that TPS window, whereas the Visa network only records a transaction. Settlement on the Visa network takes 1-3 business days.¹⁴

Given these attributes, blockchain will be pivotal in allowing agentic AI to realize its potential for consumer transactions, and the growth of agentic AI is likely to become the "killer" use case that drives blockchain adoption.

Investing in the Agentic AI Opportunity

Today, investors have positioned their portfolios to capture the AI growth opportunity by buying the stock of AI-aligned companies and related verticals, becoming limited partners of private equity funds, or investing into the energy providers and data centers that drive AI delivery. To capture the potential of agentic AI, those same portfolios should consider extending their exposure to cryptocurrencies and the alt coins being generated by blockchain-based apps and projects.

The dynamics likely to drive such adoption are outlined below.

Demand for cryptocurrencies will grow: To record a transaction on a blockchain, the AI agent will need to pay in the cryptocurrency of that underlying network. For example, to record a transaction on the Solana blockchain, the agent will need to submit SOL. As agentic payments rise, demand for the cryptocurrencies of the chains supporting such business could surge, creating value for each token holder. Initially, this will likely be driven by the machine-to-machine micropayments being generated by enterprise software systems.

Blockchain ecosystems could increase: As the number of transactions on a chain increases, and demand for that chain's cryptocurrency rises, more money flows into the chain's treasury. Blockchain foundations use this treasury money to pay grants to developers to build apps on their chain, award bug bounties¹⁵ to developers to identify security vulnerabilities and incentivize those that verify transactions for their chain. Having more capital to disperse could cause these blockchain ecosystems to grow and become ever more secure, encouraging more developers to build on these networks and issue their own alt coins to fund their projects and share ownership in their apps.

Web3 apps will take market share from Web2 apps: As more apps are deployed and more development talent is drawn to the chains, the advantages that Web3 apps offer over Web2 apps will become clear. This is already occurring for Web3 gaming apps. The gaming industry is shifting from Web2 standalone player offerings to Web3 player-owned economies. Tap-to-earn apps have drawn hundreds of millions of users worldwide. Players can now trade, buy and sell in-game items (NFTs) on secondary markets and across platforms, allowing players to verifiably own and monetize their digital game progression. Similar evolution of the user experience and ownership model could occur across an entire range of consumer apps, and this will drive interest in the alt coins being issued by these projects.

A flywheel effect can emerge: Protocols to embed AI agentic payments into blockchain-based apps already exist and could create a flywheel effect for these new issuances. Users will be able to instruct their agents to handle the payments for their transactions, eliminating the need for an individual to set up a wallet, and purchase and manage alt coins and cryptocurrencies. To the user, the experience of working with a Web3 app will look and feel no different than a Web2 offering, but they will be able to obtain more financial benefit by working within the Web3 ecosystem where the tokens provide both utility and ownership.

In a sense, the coming shift will look like the progression from (Web1) web servers and static websites to the (Web2) adoption of cloud-based businesses and interactive apps. In both instances, the underlying technology providers as well as the businesses built on those rails rotated from existing leaders to a new set of providers that drove growth for that era. Similarly, blockchains and decentralized apps/projects will be the drivers of the next transition.

At present, investors are still uncertain about the dynamics of how to capture the value being created by blockchains and their related ecosystems. They are used to centralized corporate-driven commerce. To capture the value being created by a company, they buy the equity of that company. I believe what will become increasingly clear in coming years is that in order to capture the value of decentralized networks and businesses, investors will need to buy the cryptocurrencies and alt coins issued by those entities. Such investments are likely to become key holdings in portfolios, especially for those looking to capture the emerging agentic AI opportunity.

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Endnotes

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Bug bounties are monetary rewards offered to blockchain developers to help identify security vulnerabilities on blockchain and web3 applications as an incentivized way to help keep the network secure.

WHAT ARE THE RISKS?

All investments involve risks, including possible loss of principal.

Blockchain and cryptocurrency investments are subject to various risks, including inability to develop digital asset applications or to capitalize on those applications, theft, loss, or destruction of cryptographic keys, the possibility that digital asset technologies may never be fully implemented, cybersecurity risk, conflicting intellectual property claims, and inconsistent and changing regulations. Speculative trading in bitcoins and other forms of cryptocurrencies, many of which have exhibited extreme price volatility, carries significant risk; an investor can lose the entire amount of their investment. Blockchain technology is a new and relatively untested technology and may never be implemented to a scale that provides identifiable benefits. If a cryptocurrency is deemed a security, it may be deemed to violate federal securities laws. There may be a limited or no secondary market for cryptocurrencies.

Stablecoins represent a new and rapidly evolving technological development which may be subject to regulatory uncertainty, competitive pressures, security risk, limited performance history and potential volatility. There may be risks associated with the issuance, redemption, transfer, custody, and record keeping of tokens maintained and recorded primarily on a blockchain.

Digital assets are subject to risks relating to immature and rapidly developing technology, security vulnerabilities of this technology, (such as theft, loss, or destruction of cryptographic keys), conflicting intellectual property claims, credit risk of digital asset exchanges, regulatory uncertainty, high volatility in their value/price, unclear acceptance by users and global marketplaces, and manipulation or fraud. Portfolio managers, service providers to the portfolios and other market participants increasingly depend on complex information technology and communications systems to conduct business functions. These systems are subject to a number of different threats or risks that could adversely affect the portfolio and their investors, despite the efforts of the portfolio managers and service providers to adopt technologies, processes and practices intended to mitigate these risks and protect the security of their computer systems, software, networks and other technology assets, as well as the confidentiality, integrity and availability of information belonging to the portfolios and their investors.

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